MATH 11 — ELEMENTARY STATISTICS

§6.4

Applications of Normal Distributions

Any mean, any standard deviation

Today's Goal

After completing this section, you should be able to:

- Convert values from any normal distribution to z-scores
- Calculate probabilities for non-standard normal distributions
- Find values from known probabilities for normal distributions
- Apply normal distributions to real-world problems

Learning Target 10

LT.10 Normal Probability Distributions

I understand what a normal probability distribution is. I can distinguish the difference between various normal probability distributions. I can use the standard normal distribution appropriately to calculate the probability of an event and to determine appropriate critical values.

Heights of Women

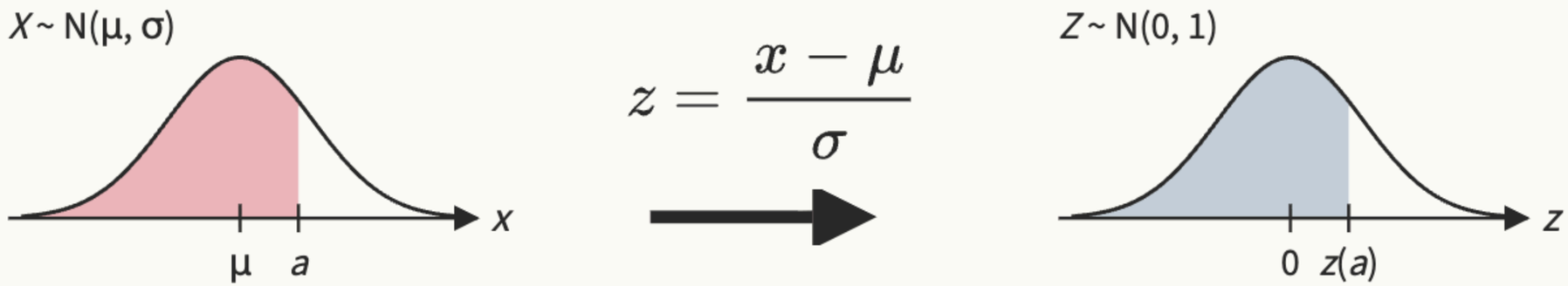
Heights of women are normally distributed with a mean of 64.2 inches and a standard deviation of 2.6 inches.

What percentage of women are taller than 68 inches?

In the previous section, we worked exclusively with the standard normal distribution where $\mu = 0$ and $\sigma = 1$. In this section, we apply these techniques to normal distributions with any mean and standard deviation.

Converting to z-scores

When working with a normal distribution that is not standard (so $\mu \neq 0$ or $\sigma \neq 1$), convert to a z-score. If $X \sim N(\mu, \sigma)$, this transformation produces $Z \sim N(0, 1)$.



The left-tail probability $P(X < a)$ equals $P(Z < z(a))$, where

$$z(a) = \frac{a - \mu}{\sigma}.$$

The same correspondence holds for the other region types.

Using Excel for Any Normal Distribution

Procedure 6.4.2 Using Excel for Any Normal Distribution

The Excel function `NORM.DIST` works directly with any normal distribution:

```
=NORM.DIST(x, mean, standard_dev, TRUE)
```

where

- `x` = the value of x you are looking up;
- `mean` = the mean μ of the distribution;
- `standard_dev` = the standard deviation σ;
- `TRUE` = cumulative probability (always use `TRUE`).

When using Excel, there is no need to manually convert to z-scores—Excel handles the conversion internally.

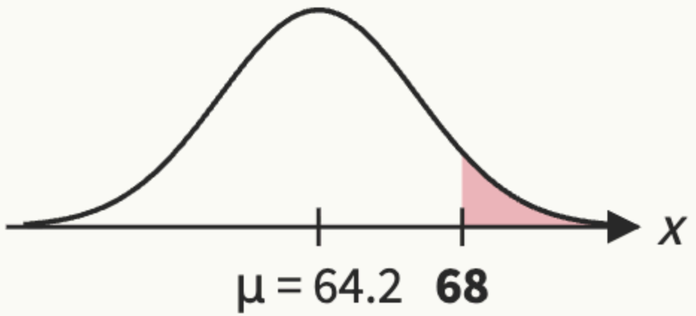
Taller Than 68 Inches

Heights of women are normally distributed with a mean of 64.2 inches and a standard deviation of 2.6 inches. Find the percentage of women who are taller than 68 inches.

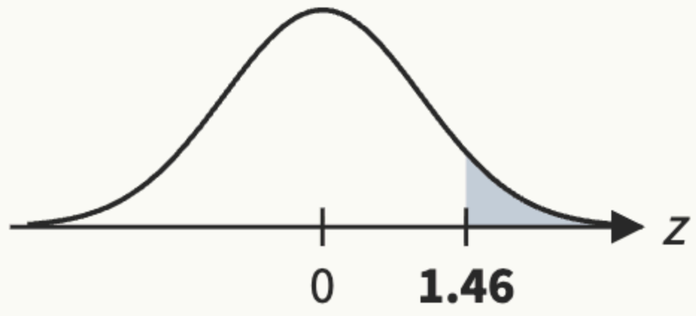
Let X represent the height of a randomly selected woman. We are being asked to find $P(X > 68)$. Since $X \sim N(64.2, 2.6)$, we will convert to the standard normal distribution. This gives us a z-score of

$$z = \frac{68 - 64.2}{2.6} \approx 1.46.$$

Taller Than 68 Inches



$$z = \frac{68 - 64.2}{2.6} \approx 1.46$$



Using Table A-1, we find that $P(z > 1.46) = 1 - 0.9279 = 0.0721$. So $P(X > 68) \approx 0.0721$.

Alternatively, we could use Excel to find $P(X > 68)$ by entering `=1-NORM.DIST(68,64.2,2.6,TRUE)` into any empty cell. This gives us a more accurate value of 0.0719.

About 7% of women are taller than 68 inches.

Converting from z-scores to x-values

Sometimes we work in the reverse direction: given a probability, find the corresponding value of x. This involves finding the z-score first and then converting back to the original scale.

Theorem 6.4.4 Converting from z-scores to x-values

If z is the z-score corresponding to a value x in a normal distribution with mean μ and standard deviation σ, then

$$x = \mu + z \cdot \sigma.$$

Using Excel to Find Values

Procedure 6.4.5 Using Excel to Find Values from Probabilities

The Excel function `NORM.INV` finds values directly for any normal distribution:

```
=NORM.INV(probability, mean, standard_dev)
```

where

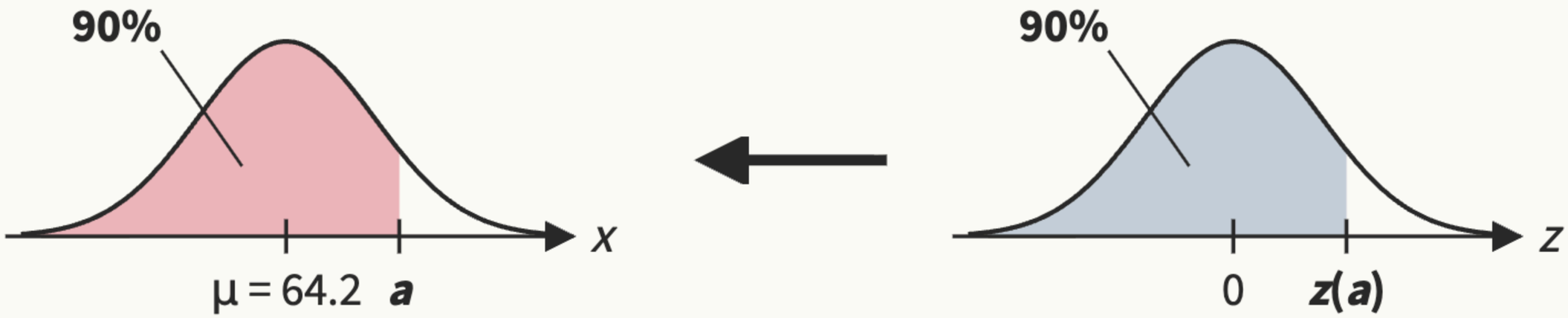
- `probability` = the cumulative probability (area to the left);
- `mean` = the mean μ of the distribution;
- `standard_dev` = the standard deviation σ.

Excel returns the x-value directly—no need to convert from z-scores.

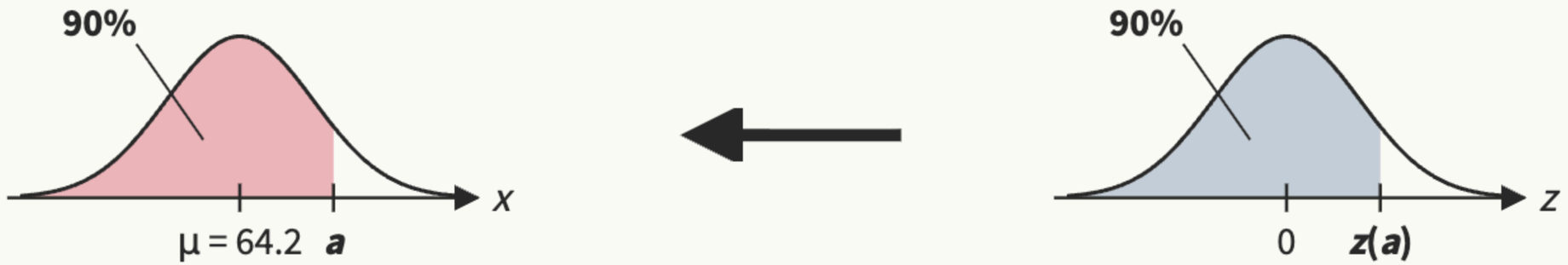
Shorter Than 90% of Women

Heights of women are normally distributed with a mean of 64.2 inches and a standard deviation of 2.6 inches. Find the height that 90% of women are shorter than.

Let X represent the height of a randomly selected woman, so $X \sim N(64.2, 2.6)$. We want a with $P(X < a) = 0.90$. First find the z-score $z(a)$ with $P(Z < z(a)) = 0.90$, then convert back.



Shorter Than 90% of Women



The value $z(a)$ is exactly $z_{0.10}$, the critical value for an area of 0.10 to the right.

From Table A-1, $z_{0.10} \approx 1.28$. Convert back to a:

$$\begin{aligned} a &= \mu + z(a) \cdot \sigma \\ &= 64.2 + 1.28 \cdot 2.6 \\ &\approx 67.528. \end{aligned}$$

So 90% of women are shorter than about 67.5 inches. Excel:

`=NORM.INV(0.90,64.2,2.6)` returns 67.53203, which rounds to 67.5 inches.

IQ Scores

IQ scores are normally distributed with $\mu = 100$ and $\sigma = 15$.

(a) What percentage of people have IQ above 130?

$P(X > 130) \approx 0.0228$**, so about** 2.28%

(b) What IQ score separates the top 10%?

About 119.2

(c) Find the range of "typical" IQ scores (middle 95%).

About 70.6 **to** 129.4